

Tutorial 2

Mofreh Saleh, Ph.D., P.E., F.ASCE

Selection of Aggregate Structure

- For a 12.5 million ESA, select a 19 mm Superpave aggregate Structure based on the following data

	Blend 1	Blend 2	Blend 3
G_{mb}	2.457	2.441	2.477
G_{mm}	2.598	2.558	2.664
G_b	1.025	1.025	1.025
P_b	5.9	5.7	6.2
P_s	94.1	94.3	93.8
P_d	4.5	4.5	4.5
G_{sb}	2.692	2.688	2.665
H_{ini}	125	133	125
H_{des}	115	118	115

Formulae

- Adjusting volumetric properties to VTM=4.0%

$$P_{be} = P_{bt} - 0.4 * (4 - VTM_t)$$

$$VMA_{est} = VMA_t + C * (4 - VTM_t)$$

C=0.1 VTM<4.0
C=0.2 VTM_t≥4.0

$$VFB_{est} = 100 \frac{(VMA_{est} - 4.0)}{VMA_{est}}$$

$$\%Gmm_{Nini_{est}} = \%G_{mmNini} - (4 - VTM_t)$$

$$P_{be_{est}} = P_{b,est} - \frac{P_s * P_{ba}}{100}$$

$$D / B_{est} = \frac{P_D}{P_{be,est}}$$

Superpave Design Criteria

VTM	4.0%				
D/B	0.6-1.2				
%TSR	80.0%				
% VMA requirements for different Max Nominal size					
Size	37.5	25	19	12.5	9.5
%VMA	11%	12%	13%	14%	15%
% Gmm Requirements at					
ESA in Millions	Nini ≤	N Des =	Nmax ≤	VFB	
<0.3	91.5	96	98	70-80	
0.3-3	90.5	96	98	65-78	
3-10	89.0	96	98	65-75	
10-30	89.0	96	98	65-75	

Superpave Mix Design



If the max Nominal Size is 19 mm and the design traffic is 20×10^6 ESA, Determine the optimum Binder Content and check All Superpave mix design criteria

P_b	5.4	5.9	6.4	6.9
G_{mb}	2.351	2.441	2.455	2.469
G_{mm}	2.57	2.558	2.53	2.51
G_b	1.025	1.025	1.025	1.025
P_s	94.6	94.1	93.6	93.1
P_d	4.5	4.5	4.5	4.5
G_{sb}	2.688	2.688	2.688	2.688
H_{ini}	125	131	126	130
H_{des}	115	118	114	112